Chapter 13

Quantitative Proteomics Foundations and Dynamic Range

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Abstract

Quantitative proteomics infers protein abundances from digested peptide intensities, propagating filtering thresholds and missingness across hierarchical levels. Direct protein quantification is physically impossible in bottom-up mass spectrometry; peptides must be identified, quantified, and aggregated into protein groups. The choice of quantification methodology determines dynamic range, throughput, and ratio accuracy: label-free quantification (LFQ) spans four to five orders of dynamic range at the cost of run-to-run stochastic missingness, whereas isobaric labeling (TMT) achieves multiplexed throughput at the cost of precursor co-isolation ratio compression. The `QFeatures` *container formalizes this hierarchy as a directed acyclic graph linking peptide-spectrum matches (PSMs), peptides, and proteins while preserving coordinate alignment and assay provenance. When summarizing peptide intensities into protein abundances, standard median aggregation is susceptible to outlying aberrant peptides, whereas M-estimation (*`robustSummary`*) downweights outlier residuals through Tukey biweight estimators. This chapter presents the mathematical foundations of MS quantitation, ratio compression kinetics, hierarchical* `QFeatures` *construction, median versus robust aggregation benchmarks on the CPTAC study, and linear modeling in R 4.6.1.*

Keywords: Quantitative proteomics, QFeatures, Dynamic range, Ratio compression, Feature aggregation, Robust summarization, CPTAC benchmark.

Chapter Overview and Architectural Roadmap

Table 13.1 compares the four primary mass spectrometry quantification paradigms and summarizes the structural requirements of hierarchical protein summarization.

Methodological Note

Tutor Warning: Aggregation Sequence Protocol

Aggregating peptides into protein groups *before* applying data filtering bakes noise and missingness

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Table 13.1: Comparison of quantitative mass spectrometry paradigms and computational trade-offs.

Methodology	Measurement Domain	Dynamic Range & Accuracy	Primary Failure Mode
1. Label-Free (LFQ)	MS1 precursor peak integration (AUC)	High dynamic range (10^4--10^5); no ratio compression	Stochastic missingness across LC-MS runs
2. Isobaric (TMT)	MS2/MS3 reporter ion intensities	High multiplexing (18-plex); high run-to-run completeness	Ratio compression from precursor co-isolation
3. DIA / SWATH	Multiplexed MS2 fragment traces	High reproducibility; wide dynamic range	Spectral deconvolution complexity
4. SILAC	MS1 isotopic doublet pairs	Direct light/heavy ratio; high accuracy	Incomplete metabolic labeling; 3-state limit

directly into protein estimates. Peptides originating from keratin contaminants, one-hit-wonder identifications, or features with $> 50\%$ missingness distort protein summaries if included during aggregation. Always filter low-quality features at the peptide tier prior to invoking `aggregateFeatures()`.

13.1 Quantitative Proteomics Methodologies and Dynamic Range

In bottom-up proteomics, proteins are proteolytically digested into peptides prior to mass spectrometric analysis. The observed signal for peptide j belonging to protein p in sample k follows a linear mixed model:

$$y_{pjk} = \mu_p + \alpha_k + \beta_j + (\alpha\beta)_{jk} + \varepsilon_{pjk} \tag{13.1}$$

where μ_p is the baseline protein abundance, α_k is the sample effect, β_j represents peptide-specific ionization efficiency, and $\varepsilon_{pjk} \sim \mathcal{N}(0, \sigma^2)$ is random measurement error.

13.1.1 Dynamic Range and Ratio Compression Kinetics

Proteomes span over six orders of magnitude in dynamic range in human plasma (10^0 to 10^{10} pg/mL). In isobaric labeling (TMT), precursor ions are selected within an isolation window (typically 0.7 to 1.2 Th). Co-eluting background peptides within this window are co-fragmented simultaneously, contributing reporter ions that distort fold-change estimates.

Let $R_{\text{true}} = I_1/I_2$ denote the true ratio of an analyte between two channels. If co-isolated interfering ions contribute background intensity with ratio $1:1$ at relative proportion $\beta = I_{\text{interfere}}/I_{\text{target}}$, the observed ratio compresses toward unity:

$$R_{\text{obs}} = \frac{I_1 + \beta I_{\text{target}}}{I_2 + \beta I_{\text{target}}} = \frac{R_{\text{true}} + \beta}{1 + \beta} \tag{13.2}$$

The percentage ratio compression is:

$$\text{Compression}(\%) = \left(1 - \frac{R_{\text{obs}} - 1}{R_{\text{true}} - 1}\right) \times 100 = \left(\frac{\beta}{1+\beta}\right) \times 100 \tag{13.3}$$

When interference $\beta = 0.50$, an actual 4.0-fold change compresses to $R_{\text{obs}} = (4 + 0.5)/(1 + 0.5) = 3.00$, representing a 33.3% compression of biological fold-change.

13.2 The QFeatures Architecture: Hierarchical Relational Graphs

The `QFeatures` container extends `MultiAssayExperiment` to manage the hierarchical multi-tier structure of mass spectrometry data. It coordinates three interconnected components:

1. **Assays**: An ordered list of `SummarizedExperiment` objects representing discrete processing tiers: `peptides` $\rightarrow$ `norm_peptides` $\rightarrow$ `proteins`.
2. **AssayLinks**: Directed acyclic graph edges linking row indices between assays, generated during aggregation or transformation via `addAssayLink()`.
3. **colData**: Experiment-wide phenotypic sample metadata shared across all assay tiers.

13.2.1 Protein Aggregation Mechanics

Converting peptide matrices into protein matrices requires row-wise summarization. Two primary algorithms are used:

1. **Median Aggregation (`colMedians`)**: Computes the sample-wise median of all peptides assigned to protein p:
$$\hat{Y}_{pk}^{\text{median}} = \text{median}_{j \in \mathscr{P}_p}(y_{pjk}) \tag{13.4}$$
While resistant to extreme values, the median discards magnitude information and can fluctuate when peptide counts are small ($|\mathscr{P}_p| \leq 2$).
2. **Robust Summarization (`robustSummary`)**: Fits a robust linear model using Tukey's biweight M-estimator:
$$\sum_{j \in \mathscr{P}_p} \psi\left(\frac{y_{pjk} - \hat{\mu}_{pk} - \hat{\beta}_j}{\hat{\sigma}_p}\right) = 0 \tag{13.5}$$
where $\psi(u) = u(1-(u/c)^2)^2$ for $|u| \leq c$ ($c = 4.685$) and zero otherwise. Peptides with large residual deviations are downweighted iteratively, preventing single aberrant peptides from distorting protein quantification.

13.3 CPTAC Benchmark Analysis and Differential Testing

The Clinical Proteomic Tumor Analysis Consortium (CPTAC) study design evaluates quantitative pipelines using yeast spike-in standards across two dilution conditions (6A and 6B) with three replicates each. Listing 4.1 in Chapter 13 loads the genuine 11,466-peptide MaxQuant identification table from `MsDataHub`.

Listing 13.1 ingests the CPTAC dataset, filters missing values, and normalizes intensities.

R Lab Listing 1: Listing 13.1: Ingestion and Normalization of the CPTAC Benchmark [R 4.4 / Bioconductor]

```
library(QFeatures)
library(SummarizedExperiment)
library(MsDataHub)

peptides_file <- MsDataHub::cptac_a_b_peptides.txt()
```

```
cptac_raw <- read.delim(peptides_file, check.names = FALSE)

intensity_cols <- grep("^Intensity 6[AB]_", colnames(cptac_raw), value = TRUE)
sample_info <- DataFrame(
  sample    = intensity_cols,
  condition = sub("Intensity (6[AB])_.*", "\\1", intensity_cols),
  replicate = sub("Intensity 6[AB]_(.*)", "\\1", intensity_cols),
  row.names = intensity_cols
)

peptide_data <- as.matrix(cptac_raw[, intensity_cols])
rownames(peptide_data) <- make.unique(cptac_raw$Sequence)
peptide_data[peptide_data == 0] <- NA_real_

cptac_se <- SummarizedExperiment(
  assays  = list(peptides = peptide_data),
  rowData = DataFrame(Proteins = cptac_raw$Proteins, row.names = rownames(peptide_data)),
  colData = sample_info
)

cptac_qf <- QFeatures(list(peptides = cptac_se))
cptac_qf <- logTransform(cptac_qf, i = "peptides", name = "log_peptides")
cptac_qf <- normalize(cptac_qf, i = "log_peptides", name = "norm_peptides",
                      method = "center.median")

# Filter features with > 50% missingness and empty protein annotations
cptac_clean <- filterNA(cptac_qf, i = "norm_peptides", pNA = 0.5)
has_prot <- rowData(cptac_clean[["norm_peptides"]])$Proteins != ""
cptac_clean <- cptac_clean[has_prot, , ]

cat(sprintf("Raw CPTAC Peptides: %d x %d samples\n",
            nrow(cptac_qf[["peptides"]]), ncol(cptac_qf[["peptides"]])))
cat(sprintf("Retained Peptides (pNA <= 0.5): %d\n", nrow(cptac_clean[["norm_peptides"]])))
```

```
Raw CPTAC Peptides: 11466 x 6 samples
Retained Peptides (pNA <= 0.5): 6643
```

Of 11,466 raw peptides, 45.25% of matrix cells contain missing values. Filtering for pNA ≤ 0.5 and removing 7 peptides lacking protein annotations yields 6,643 analytical peptides (58.0% retention).

13.4 Median vs RobustSummarization Benchmark

Listing 13.2 aggregates the 6,643 peptides to 1,451 protein groups using both `colMedians` and `MsCoreUtils::robustSummary`, followed by differential abundance testing with `limma`.

R Lab Listing 2: Listing 13.2: Protein Aggregation and Differential Abundance Testing [R 4.4 / Bioconductor]

```
library(MsCoreUtils)
library(limma)
library(dplyr)

# Aggregation 1: Column medians
```

```
cptac_clean <- aggregateFeatures(cptac_clean, i = "norm_peptides", fcol = "Proteins",
                                 name = "proteins_median", fun = colMedians, na.rm = TRUE)

# Aggregation 2: Robust M-estimation
cptac_clean <- aggregateFeatures(cptac_clean, i = "norm_peptides", fcol = "Proteins",
                                 name = "proteins_robust", fun = MsCoreUtils::robustSummary, na.rm = TRUE)

# Differential abundance helper
test_diff <- function(qf, assay_name) {
  se_sub <- filterNA(getWithColData(qf, assay_name), pNA = 0)
  design <- model.matrix(~ condition, data = colData(se_sub))
  fit <- eBayes(lmFit(assay(se_sub), design))
  res <- topTable(fit, coef = 2, number = Inf)
  res$protein <- rownames(res)
  res
}

res_med <- test_diff(cptac_clean, "proteins_median")
res_rob <- test_diff(cptac_clean, "proteins_robust")

sig_med <- res_med %>% filter(adj.P.Val <= 0.05, abs(logFC) >= 1.0)
sig_rob <- res_rob %>% filter(adj.P.Val <= 0.05, abs(logFC) >= 1.0)
shared  <- intersect(sig_med$protein, sig_rob$protein)
jaccard <- length(shared) / length(union(sig_med$protein, sig_rob$protein))

cat(sprintf("Aggregated Protein Groups: %d\n", nrow(cptac_clean[["proteins_median"]])))
cat(sprintf("Significant Proteins (Median):        %d\n", nrow(sig_med)))
cat(sprintf("Significant Proteins (RobustSummary): %d\n", nrow(sig_rob)))
cat(sprintf("Shared Significant Proteins:          %d (Jaccard = %.4f)\n", length(shared), jaccard))
```

```
Aggregated Protein Groups: 1451
Significant Proteins (Median):        3
Significant Proteins (RobustSummary): 12
Shared Significant Proteins:         2 (Jaccard = 0.1538)
```

At significance thresholds $|\log_2 \mathrm{FC}| \geq 1.0$ and FDR ≤ 0.05, median aggregation identifies 3 significant proteins, whereas robust summarization identifies 12 significant proteins. Only 2 proteins are shared, yielding a Jaccard similarity index of 0.1538. Median aggregation suffers from variance inflation caused by unmodeled peptide effects, whereas robust summarization downweights aberrant peptides, stabilizing protein variance and improving statistical power.

13.5 Worked Exercises

Exercise 13.1: Aggregation Robustness and Classification Shift

Problem Statement: In the CPTAC benchmark analysis, median aggregation detected 3 significant proteins while robust summarization detected 12. Explain why 11 proteins differed in classification between the two methods.

Solution: Median aggregation takes the simple column median across all peptides mapped to a protein. If a protein is quantified by three peptides where one peptide exhibits an atypical ionization shift or co-eluting chemical interference, the sample median can shift substantially across replicates, inflating the estimated residual variance in `limma` (s_g^2).

Robust summarization fits an additive linear model ($y_{jk} = \mu_k + \beta_j + \varepsilon_{jk}$) using Tukey's biweight M-estimator. The estimator downweights individual peptide residuals $|r_{jk}| > c\hat{\sigma}$, isolating genuine protein-level changes from peptide-specific ionization noise. This reduces residual variance $\tilde{s}_g^2$, enabling 10 true yeast spike-in proteins to cross the FDR ≤ 0.05 threshold.

Exercise 13.2: Missingness Threshold Filtering Sweep

Problem Statement: Evaluating `filterNA()` on the CPTAC peptide assay at missingness thresholds pNA $=$ 0.2, 0.5, and 0.8 retained 5,049, 6,650, and 7,584 peptides, respectively. Analyze the analytical trade-offs of stringent versus permissive missingness filtering.
Solution:

1. **Stringent Filtering (pNA** $= 0.2$**)**: Retains 5,049 peptides (44.0%). Eliminates imputation uncertainty but discards peptides detected in only one biological condition, filtering out proteins completely absent in the control group.

2. **Permissive Filtering (pNA** $= 0.8$**)**: Retains 7,584 peptides (66.1%). Maximizes proteome coverage but introduces high missingness proportions that require downstream imputation algorithms (such as MinProb or KNN), risking imputation artifacts in statistical inference.

Exercise 13.3: Multi-Tier PCA Diagnostics

Problem Statement: In a quantitative proteomics study, peptide-level PCA separates samples by both injection order and biological condition, whereas protein-level PCA clusters predominantly by biological condition (6A vs 6B). State what this diagnostic pattern indicates about the origin of analytical variance.
Solution: This pattern demonstrates that chromatographic drift and electrospray ionization fluctuations operate at the peptide level. Because peptide-specific technical variations are largely independent across distinct peptide sequences, aggregating multiple peptides into protein groups averages out independent technical noise:

$$\mathrm{Var}(\bar{Y}_p) \approx \frac{\sigma^2_{\text{peptide}}}{|\mathscr{P}_p|} + \sigma^2_{\text{protein}} \tag{13.6}$$

The biological condition signal dominates at the protein tier because it affects all peptides belonging to that protein systematically.

Exercise 13.4: MSstats Converter Mapping

Problem Statement: Specify the conversion function from the `MSstatsConvert` package required to transform output from (a) MaxQuant, (b) DIA-NN, and (c) Spectronaut into standard MSstats format.
Solution:

1. **MaxQuant**: `MSstatsConvert::MaxQtoMSstatsFormat(evidence, proteinGroups, annotation)`

2. **DIA-NN**: `MSstatsConvert::DIANNtoMSstatsFormat(report, annotation)`

3. **Spectronaut**: `MSstatsConvert::SpectronauttoMSstatsFormat(report, annotation)`

The resulting standardized data frame feeds into `MSstats::dataProcess()` for run-level summarization.

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